

STUDENT HANDOUT

The $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ Equilibrium: Chemical Thermodynamics, Statistical Mechanics, Computational Chemistry and Chemical Kinetics

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Note on Updates

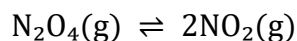
This student handout corresponds to the version included as Supporting Information with this paper. Any future updates or revisions to the handout will be made available at: https://github.com/daferro/Notebooks-Chemical_Equilibrium

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1. Introduction

35 This material provides a comprehensive guide to the analysis of the gas-phase dissociation equilibrium:



Using concepts from thermodynamics, statistical mechanics, computational chemistry, and chemical kinetics, you will:

- 40
- Investigate how the equilibrium position depends on temperature and pressure.
 - Compute equilibrium constants from first principles.
 - Examine the time-dependent approach to equilibrium through analytical solutions of the rate equations.

This integrated approach will allow you to connect molecular-level properties to macroscopic
45 observables and dynamic behavior in chemical systems.

2. Computer laboratory

A Python interactive notebook (Notebook1.ipynb) will be used in this computer laboratory to study the chemical equilibrium $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$.

2.1. Interactive Python notebook in Google Colab

When accessing Google Colab, the platform presents an interface designed to simplify the creation and management of interactive notebooks. Upon launching Colab, the user is typically greeted with a welcome screen, from which a dialog window titled “*Open notebook*” can be accessed (see Figure 1). This window centralizes all options for creating or opening notebooks.

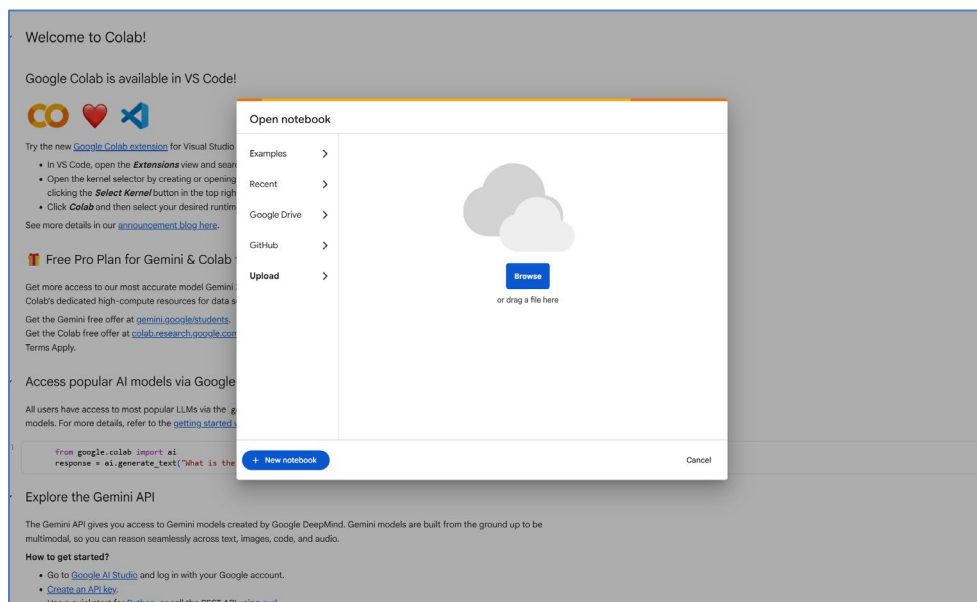


Figure 1. The “Open notebook” dialog in Google Colab, showing the available options for creating a new notebook or opening existing notebooks from examples, recent files, Google Drive, GitHub, or by uploading a local notebook file.

The *Open notebook* dialog is organized around a navigation panel on the left-hand side, which provides several sources from which a notebook can be opened:

- **Examples.** Provides a curated set of example notebooks maintained by Google. These are useful for exploring Colab’s capabilities or for instructional demonstrations.
- **Recent.** Displays notebooks that have been opened recently by the user, allowing quick access to ongoing work.

- **Google Drive.** Allows the user to browse notebooks stored in their Google Drive. This is the most common option when working on course materials or assignments saved in the cloud.
- **GitHub.** Enables opening notebooks directly from public or private GitHub repositories by specifying a repository URL. This option is often used in teaching contexts where instructors distribute notebooks via version control.
- **Upload.** Permits uploading a local notebook file (.ipynb) from the user's computer. Files can be selected using the *Browse* button or by dragging and dropping them into the dialog.

Alternatively, the notebook can be accessed directly from the GitHub repository: https://github.com/daferro/Notebooks-Chemical_Equilibrium. On the repository's main page, the README section contains badges that allow the Python notebooks to be opened and executed in Google Colab with a single click. Use of Google Colab requires a valid Google account.

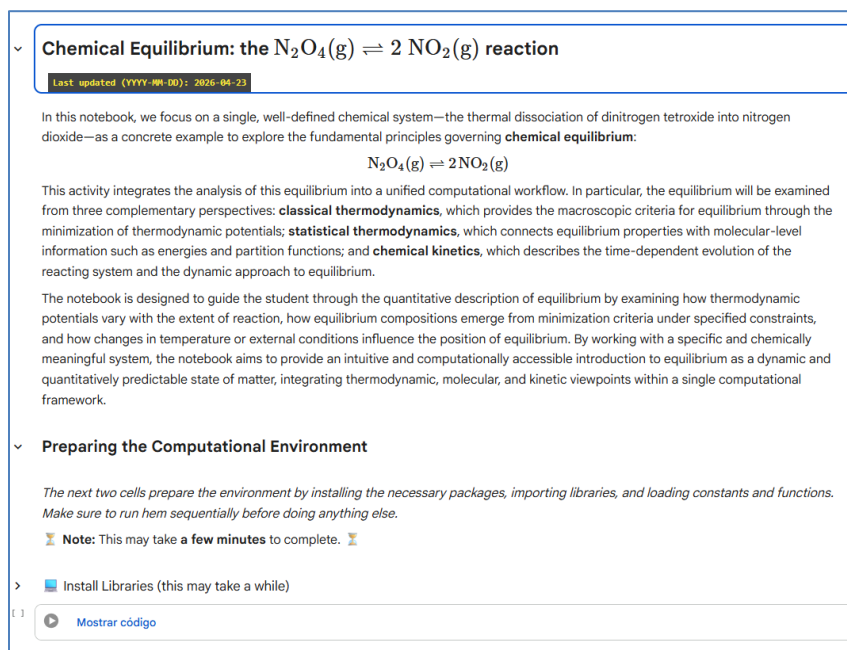


Figure 2. Standard Colab interface for interactive Notebook 1

After a notebook is opened, the following features and steps are relevant:

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- 80 1. The notebook interface appears (see Figure 2), composed of *cells* that can contain either code or formatted text (*Markdown*).
2. By default, Colab assigns a virtual execution environment (runtime). If required, the runtime type (e.g., CPU, GPU) can be modified via the *Runtime* menu.
3. Any changes made to notebooks stored in Google Drive are saved automatically.
- 85 4. The notebooks are organized into collapsible sections. Before running any analysis, the section titled “Preparing the Computational Environment” must be executed; it contains the setup cells that install required packages, import libraries, and load constants and functions. These cells can be expanded and executed using the play button on the left of each cell.
- 90 5. Once the environment is prepared, the remaining sections can be expanded and executed in order. Code cells are run individually by clicking the play button or using the keyboard shortcut *Shift + Enter*.

3. Chemical thermodynamics perspective

3.1. Gibbs energy of reaction

95 Consider a chemical reaction involving ideal gases, written in terms of stoichiometric coefficients ν_i (positive for products and negative for reactants). The *Gibbs energy of reaction* is defined as

$$\Delta_r G = \sum_i \nu_i \mu_i, \quad (1)$$

where μ_i is the *chemical potential* of species i . This quantity, $\Delta_r G$, tells us whether the reaction
100 will proceed spontaneously under given conditions.

3.2. Chemical potential of an ideal gas

For an ideal gas, the chemical potential depends on temperature and partial pressure according to

$$\mu_i(T, p_i) = \mu_i^\circ(T) + RT \ln \left(\frac{p_i}{p^\circ} \right), \quad (2)$$

105 where p_i is the partial pressure of species i , $p^\circ = 1$ bar is the standard pressure, $\mu_i^\circ(T)$ is the *standard chemical potential*, and R is gas constant. The logarithmic term accounts for the effect of composition on the chemical potential.

3.3. Equilibrium condition and equilibrium constant

At *chemical equilibrium*, the Gibbs energy of reaction is zero ($\Delta_r G = 0$), which gives

$$\Delta_r G^\circ(T) = -RT \ln \left[\prod_i \left(\frac{p_{i,eq}}{p^\circ} \right)^{\nu_i} \right] = -RT \ln K_p^\circ(T), \quad (3)$$

where $\Delta_r G^\circ(T)$ is the *standard Gibbs energy of reaction*, and $K_p^\circ(T)$ is the *standard pressure equilibrium constant*.

3.4. Temperature dependence of the standard Gibbs energy

The *standard Gibbs energy of reaction* at temperature T can be estimated from
115 thermodynamic data at 298 K:

$$\Delta_r G^\circ(T) = \Delta_r H^\circ(298 \text{ K}) - T \cdot \Delta_r S^\circ(298 \text{ K}) + \Delta_r C_p^\circ(298 \text{ K}) \cdot \left[T - 298 \text{ K} - T \ln \left(\frac{T}{298 \text{ K}} \right) \right]. \quad (4)$$

Here,

- ✓ $\Delta_r H^\circ$ is the standard enthalpy of reaction,
- ✓ $\Delta_r S^\circ$ is the standard entropy of reaction, and
- ✓ $\Delta_r C_p^\circ$ is the change in standard molar heat capacity at constant pressure.

3.5. Temperature dependence of the standard enthalpy and entropy

The corresponding expressions for the temperature dependence of the *standard enthalpy and entropy of reaction* are

$$\Delta_r H^\circ(T) = \Delta_r H^\circ(298 \text{ K}) + \Delta_r C_p^\circ(298 \text{ K}) \cdot (T - 298 \text{ K}), \quad (5)$$

$$\Delta_r S^\circ(T) = \Delta_r S^\circ(298 \text{ K}) + \Delta_r C_p^\circ(298 \text{ K}) \cdot \ln \left(\frac{T}{298 \text{ K}} \right). \quad (6)$$

In eqs (4)-(6), the change in standard molar heat capacity at constant pressure $\Delta_r C_p^\circ(298 \text{ K}) = \sum_i \nu_i C_{p,m,i}^\circ(298 \text{ K})$ is assumed to be temperature independent. This approximation is commonly used and is sufficiently accurate over moderate temperature ranges.

3.6. Gibbs-Helmholtz equation

A useful related relation connecting the Gibbs energy and enthalpy of reaction is the *Gibbs-Helmholtz equation*:

$$\frac{d(\Delta_r G^\circ/T)}{dT} = -\frac{\Delta_r H^\circ}{T^2}. \quad (7)$$

This equation highlights how the temperature dependence of the Gibbs energy is governed by the enthalpy of reaction.

3.7. Equilibrium composition and extent of reaction

At *constant temperature and pressure*, the equilibrium composition can be determined by *minimizing the total Gibbs energy* of the system. Introducing the *extent of reaction* ξ , the Gibbs energy can be written as

$$G(\xi) = \xi \cdot \Delta_r G^* + \Delta_{\text{mix}} G(\xi) - \Delta_{\text{mix}} G(0), \quad (8)$$

140 where $\Delta_r G^* = \sum_i \nu_i \mu_i^*$ is the Gibbs energy change for converting pure reactants into pure products at the given pressure, and $\Delta_{\text{mix}} G(\xi)$ accounts for the *Gibbs energy change due to mixing*. For simplicity, it is assumed that $G(0) = 0$ in equation (8).

At equilibrium, the Gibbs energy is minimized with respect to ξ , which is equivalent to the condition $\Delta_r G = (\partial G / \partial \xi)_{T,p} = 0$ at $\xi = \xi_{\text{eq}}$.

145 If the chemical reaction takes place at *constant temperature and volume*, the appropriate thermodynamic potential to be minimized is the *Helmholtz energy*. In this case, the Helmholtz energy as a function of the extent of reaction can be written as

$$A(\xi) = \xi \cdot \Delta_r G^\circ + RT \cdot \sum_i \left[n_i(\xi) \cdot \ln \left(\frac{p_i(\xi)}{e \cdot p^\circ} \right) - n_i(0) \cdot \ln \left(\frac{p_i(0)}{e \cdot p^\circ} \right) \right], \quad (9)$$

where we assume that $A(0) = 0$.

150 3.8. Two-component systems: the intercept method

For a chemical equilibrium involving only two species, denoted A and B, the *molar Gibbs energy* of the mixture can be expressed in terms of the chemical potentials as:

$$G_m = \mu_A + (\mu_B - \mu_A) \cdot y_B, \quad (10)$$

where y_B is the mole fraction of species B.

155 The *intercept method for partial molar quantities* provides a convenient graphical way to determine the chemical potentials at a given composition.

As illustrated in Figure 3, the tangent line to the G_m curve at a given composition $y_B = z$ plays a central role. The intercepts of this tangent line (shown in red in the figure) have a clear interpretation:

- 160 ✓ The intercept at $y_B = 0$ corresponds to the chemical potential of A at composition z , denoted $\mu_{A,z}$;

- ✓ The intercept at $y_B = 1$ corresponds to the chemical potential of B at the same composition, $\mu_{B,z}$.

Thus, a single tangent line simultaneously yields the chemical potentials of both components at the chosen composition.

At equilibrium, the reaction Gibbs energy satisfies the condition $\Delta_r G = \nu_A \mu_A + \nu_B \mu_B = 0$. The composition of the system can therefore be adjusted until this relationship is satisfied, thereby defining the equilibrium mole fraction.

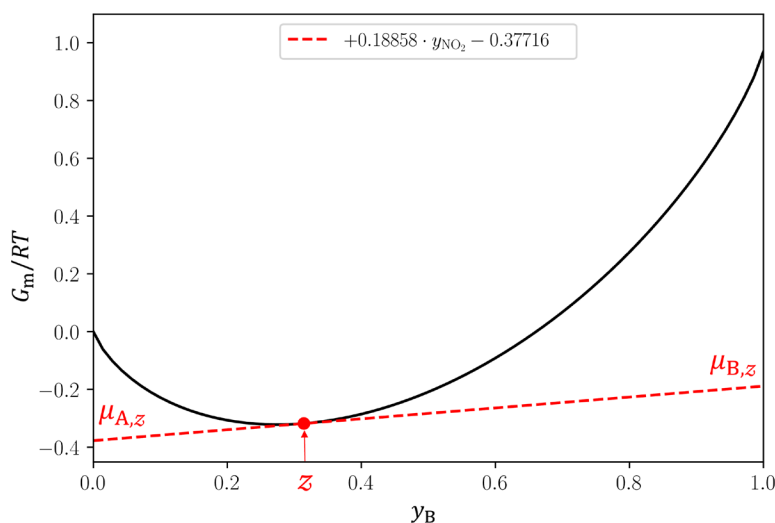


Figure 3. Variation of G_m/RT with the mole fraction of B (NO_2) for the reaction $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ at $T = 298 \text{ K}$ and $p = 1 \text{ bar}$, assuming an initial mixture containing 1 mol of reactant. The red line represents the tangent to the G_m curve.

The thermodynamic data required for the equilibrium $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ can be taken from Atkins.¹

4. Statistical mechanics perspective

4.1. The partition function

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Statistical mechanics explains thermodynamic quantities in terms of molecular energy levels. Molecules possess translational, rotational, vibrational, and electronic energy states, whose populations are governed by Boltzmann statistics. All relevant molecular information is collected in the *partition function*.

A larger partition function corresponds to more accessible energy states, leading to higher entropy and lower Gibbs energy. Thus, the partition function provides a microscopic connection between molecular structure and macroscopic thermodynamics.

From statistical mechanics, the standard chemical potential is given by

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$$\mu^\circ = E_{0,m} - RT \ln \left(\frac{q_m^\circ}{N_A} \right), \quad (11)$$

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where $E_{0,m}$ is the molar ground-state electronic energy, including zero-point vibrational energy (ZPVE), q_m° is the standard molar partition function, and N_A is Avogadro's number. The first term represents the minimum molecular energy, while the logarithmic term accounts for thermally accessible states. As temperature increases, more states are populated, increasing the partition function and lowering the chemical potential.

The total partition function can be written as a product of contributions from translational, rotational, vibrational, and electronic degrees of freedom.

4.2. Electronic structure calculations

To calculate standard enthalpies, entropies, and Gibbs energies using statistical mechanics, reliable molecular properties such as ground-state energies, rotational constants, and vibrational frequencies are required. These quantities are obtained from optimized molecular geometries using *quantum chemical calculations*, which you have already encountered in the quantum chemistry course in the fall term. As illustrated in Figure 4, a molecular geometry optimization systematically adjusts the atomic positions, starting from an initial structure, progressively lowering the calculated electronic energy until a stable geometry corresponding to a minimum on the potential energy surface is reached.

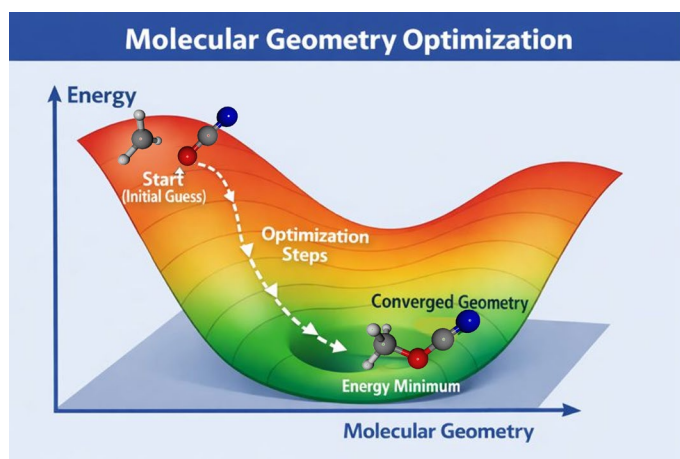


Figure 4. Schematic representation of a geometry optimization on a potential energy surface, illustrating how an initial molecular structure is iteratively refined through successive optimization steps until a converged geometry corresponding to a local energy minimum is reached.

In this laboratory, calculations are performed using the PySCF² quantum chemistry package. You will optimize the geometries of N_2O_4 and NO_2 using Density Functional Theory (DFT).

From the optimized structures, rotational constants were extracted directly. Vibrational frequencies are then calculated by evaluating the Hessian matrix, which provides the information needed for vibrational contributions to thermodynamic properties, confirms that the optimized structures correspond to true energy minima through the absence of imaginary frequencies, and allows the calculation of the ZPVE. The resulting electronic energies including the ZPVE correspond to $E_{0,m}$ in equation (11).

You will perform calculations using three hybrid DFT functionals: B3LYP, O3LYP, and PBE0, to assess the sensitivity of the predicted equilibrium to the electronic structure method.

4.3. Understanding the Role of $\Delta_r C_p^\circ$ in equation (4)

The temperature dependence of $\Delta_r G^\circ$ (equation 4) depends not only on enthalpy and entropy, but also on the change in heat capacity, $\Delta_r C_p^\circ$. While $\Delta_r C_p^\circ$ is often taken from experimental tables, it can also be estimated using simple physical ideas.

A basic approach uses the *classical equipartition principle*, which counts the contributions of *active* rotational (n^R) and vibrational (n^{V^*}) modes to the molar heat capacity

$$C_{p,m}^\circ = \left[1 + \frac{1}{2} (3 + n^R + 2 \cdot n^{V^*}) \right] \cdot R. \quad (12)$$

Rotational modes are usually fully active at room temperature, but many vibrational modes have much higher energy spacings and contribute little, so they can initially be neglected.

A more accurate estimate accounts for partially active vibrational modes. This can be done either by fitting equation (4) to the full statistical-mechanical results, with $\Delta_r C_p^\circ$ treated as a fitting parameter, or by calculating n^{V^*} directly from the vibrational frequencies of the molecule, as shown in Figure 5. This figure illustrates that low-frequency vibrational modes contribute substantially more to the heat capacity than high-frequency modes.

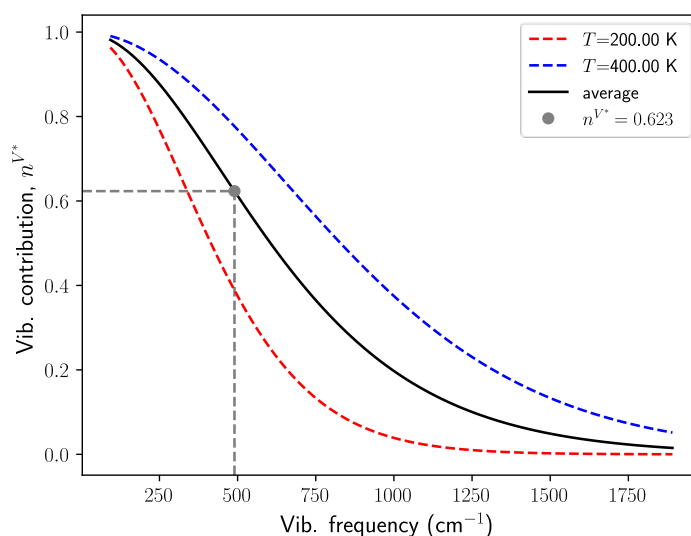


Figure 5. Contribution of individual vibrational modes to the heat capacity according to statistical mechanics

The trends shown in Figure 5 can be readily understood by examining Figure 6. It qualitatively depicts the energy levels of low- and high-frequency harmonic oscillators, along with their relative populations at a given temperature.

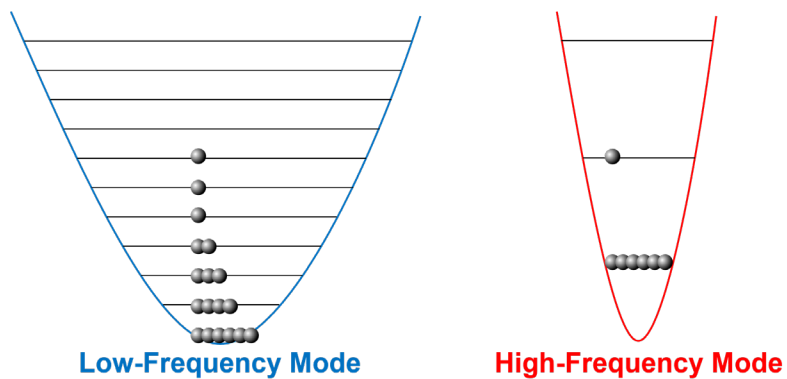


Figure 6. Low-frequency modes contribute more to the heat capacity than high-frequency modes because they are more easily excited.

5. Chemical kinetics perspective

5.1. Background

240 Consider the gas-phase reaction $A \rightleftharpoons 2B$. If we express the reaction rate in terms of the extent of reaction and the reactor volume, it can be written as

$$r = \frac{1}{V} \frac{d\xi}{dt} = k_{fw}[A] - k_{bw}[B]^2, \quad (13)$$

where k_{fw} and k_{bw} are the forward and backward (or reverse) rate coefficients, respectively.

At *chemical equilibrium*, the concentrations of all species are constant in time. Therefore,
245 $\frac{d\xi}{dt} = 0$. This means that the *forward and backward reaction rates are equal*

$$k_{fw}[A]_{eq} = k_{bw}[B]_{eq}^2. \quad (14)$$

Rearranging this expression gives the familiar relationship between kinetics and equilibrium:

$$\frac{k_{fw}}{k_{bw}} = \frac{[B]_{eq}^2}{[A]_{eq}} = K_c \quad (15)$$

where K_c is the concentration-based equilibrium constant. It is related to the standard
250 concentration equilibrium constant by: $K_c = K_c^\circ (c^\circ)^{\Delta\nu}$.

As a specific example, the forward rate constant for the dissociation of N_2O_4 can be described by an Arrhenius-type expression

$$k_{fw}(T) = A \cdot e^{-\frac{B}{T}}, \quad (16)$$

where $A = 1.15 \cdot 10^6 \text{ s}^{-1}$ and $B = 6460 \text{ K}$.³ Once $k_{fw}(T)$ is known, the backward rate constant
255 $k_{bw}(T)$ can be obtained directly using equation (15).

Finally, to solve the exercises, you may need to recall the relationship between the standard equilibrium constants expressed in terms of pressure and concentration:

$$K_p^\circ = K_c^\circ \cdot \left(\frac{c^\circ RT}{p^\circ} \right)^{\Delta\nu}, \quad (17)$$

with $c^\circ = 1 \text{ M}$.

260 6. Exercises part I (Chemical Thermodynamics)

Names:

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Exercise 1: Thermodynamic properties at 298 K. **Using the standard thermodynamic data from the table provided in Atkins' *Physical Chemistry*:**

a) Determine $\Delta_r H^\circ(298\text{ K})$, $\Delta_r S^\circ(298\text{ K})$, and $\Delta_r G^\circ(298\text{ K})$.

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b) From $\Delta_r G^\circ(298\text{ K})$, determine $K_p^\circ(298\text{ K})$.

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c) Determine $\Delta_r C_p^\circ(298\text{ K})$ and express the result in terms of the gas constant R .

Exercise 2: Temperature dependence of K_p° . **Using the notebook**, determine K_p° at:

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a) 250 K.

b) 350 K.

c) 400 K.

Questions:

1. How does temperature affect the equilibrium? Explain the trend.

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2. **Demonstrate by hand** that the Gibbs–Helmholtz relation holds assuming $\Delta_r G^\circ(T) = \Delta_r H^\circ(298\text{ K}) - T \cdot \Delta_r S^\circ(298\text{ K})$.

3. Using the assumption of the previous question, **show that** $K_p^\circ(T) = e^{\Delta_r S^\circ(298\text{ K})/R} e^{-\Delta_r H^\circ(298\text{ K})/RT}$

Exercise 3: Equilibrium at **constant p** . For the initial conditions: 1.00 mol of N_2O_4 at 1 bar and 298 K:

Questions:

1. **Calculate ξ_{eq} at 298 K manually.**

2. Compare your manual result with the result from the notebook.

3. **Using the notebook**, indicate and explain how the equilibrium constant $K_p^\circ(T)$ and the one expressed in mole fractions (K_y) vary from 1 to 2 bar?

Exercise 4: Equilibrium at **constant V** . For the initial conditions: 1.00 mol of N_2O_4 , 24.78 L and 298 K (questions 2 and 3 refer to the shift relative to constant pressure):

Questions:

1. Using the **notebook**, determine the equilibrium pressure:
2. Determine the direction in which the equilibrium shifts:
3. Explain the change using Le Châtelier's principle:

Exercise 5: Equilibrium composition using the **intercept method for a binary mixture**. Determine the equilibrium composition at 350 K and 1 bar **using the notebook**.

Questions:

1. What are the equilibrium mole fractions?

2. What are the equilibrium chemical potentials?

3. For which reaction stoichiometries involving only two species would the minimum in G_m correspond to the equilibrium point?

350 7. Exercises part II (Statistical Mechanics)

Names:

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Exercise 6: Geometry optimizations of the reactant and product. Using **the notebook**, optimize the geometries of the reactant and product with the three proposed computational methods.

1. What would it indicate if imaginary frequencies were observed in your calculations?

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2. Comparing the vibrational frequencies of both molecules, what conclusions can be drawn?

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Exercise 7: Thermodynamic Properties and equilibrium constants. Using **the notebook**, calculate $\Delta_r H^\circ$, $\Delta_r S^\circ$, $\Delta_r G^\circ$, and the equilibrium constant at 298 K using the three DFT methods.

Method	$\Delta_r H^\circ$	$\Delta_r S^\circ$	$\Delta_r G^\circ$	K_p°

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Questions:

1. Which DFT method provides the best agreement with experiment? What is the relative error in terms of the equilibrium constant K_p° ?

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Exercise 8: Understanding the role of $\Delta_r C_p^\circ$ in equation (4)

Questions:

1. Using equation (12), **calculate by hand** the value of $\Delta_r C_p^\circ$ in the classical (high-temperature) limit, where all rotational and vibrational modes are assumed to be fully active.

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2. Using equation (12), **calculate by hand** the value of $\Delta_r C_p^\circ$ assuming that vibrational contributions are entirely neglected ($n^{V^*} = 0$).

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3. **Using the notebook** to determine the contribution to n^{V^*} of a vibrational mode with a frequency of 150 cm^{-1} and another of 1500 cm^{-1} . What can you conclude from the result?

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4. Which molecule contributes more significantly to n^{V^*} ? Why?

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8. Exercises part III (Chemical Kinetics)

Names:

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Exercise 9: Forward and reverse rate coefficients

Questions:

1. Using the experimental value of the standard Gibbs energy of reaction at 298 K and equation (15), together with the experimental values of A and B , **calculate by hand** the forward and reverse rate coefficients. Include the correct units in your results

2. **Use the notebook** to vary the parameters A and B in equation (16) and determine their influence on the temporal profiles for a constant-volume reactor. Which one has the strongest effect? Explain your observations.

(Hint: you can compare the values of k_{fw} using the baseline value of A and a value twice as large, keeping B constant, and do the analogous test for B keeping A constant.)

Exercise 10: Time Evolution toward equilibrium

Questions:

1. **Use the notebook** to estimate the time required for the system to reach equilibrium at constant pressure of 1 bar (at 298 K, starting from an initial volume of 24.78 L):

2. **Use the notebook** to estimate the time required for the system to reach equilibrium at constant volume of 24.78 L (at 298 K, starting from an initial pressure of 1 bar):

3. Explain the difference observed in the equilibration times.

4. What would you change in the initial conditions to obtain the opposite trend?

9. References

- 470 (1) Atkins, P. W.; De Paula, J.; Keeler, J. *Atkins' Physical Chemistry*, 11th ed.; Oxford University Press, London, 2018.
- (2) Sun, Q.; Zhang, X.; Banerjee, S.; Bao, P.; Barbry, M.; Blunt, N. S.; Bogdanov, N. A.; Booth, G. H.; Chen, J.; Cui, Z.-H.; et al. Recent developments in the PySCF program package. *J. Chem. Phys.* **2020**, *153* (2). DOI: 10.1063/5.0006074.
- 475 (3) Atkinson, R.; Baulch, D. L.; Cox, R. A.; Crowley, J. N.; Hampson, R. F.; Hynes, R. G.; Jenkin, M. E.; Rossi, M. J.; Troe, J. Evaluated kinetic and photochemical data for atmospheric chemistry: Volume I - gas phase reactions of O_x, HO_x, NO_x and SO_x species. *Atmos. Chem. Phys.* **2004**, *4* (6), 1461-1738. DOI: 10.5194/acp-4-1461-2004.